



May 2025

(For CS)

Max. Marks: 100

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

20 Marks

- 1-

5. a) Define the Perceptron Model. Explain the Backpropagation algorithm and the role of the learning rate in a backpropagation network.
- b) Apply the Naïve Bayes Classifier to analyze the given car theft problem, considering attributes such as Color, Type, and Origin, with the target variable Stolen (Yes/No). Use probabilistic classification to determine whether a Red Domestic SUV is likely to be stolen.

Example No.	Color	Type	Origin	Stolen?
1	Red	Sports	Domestic	Yes
2	Red	Sports	Domestic	No
3	Red	Sports	Domestic	Yes
4	Yellow	Sports	Domestic	No
5	Yellow	Sports	Imported	Yes
6	Yellow	SUV	Imported	No
7	Yellow	SUV	Imported	Yes
8	Yellow	SUV	Domestic	No
9	Red	SUV	Imported	No
10	Red	Sports	Imported	Yes

6. a) Explain the following concepts of Ensemble Learning with example:
- Bagging
 - Boosting
- b) Summarize the Decision Tree Algorithm and explain each step in detail. Calculate the Entropy and Information Gain for the attribute "Outlook" to assess its effectiveness in partitioning the following dataset:

Day	Outlook	Temperature	Humidity	Wind	Play Tennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

Unit – III

7. a) Apply the k-means clustering algorithm to group a given set of points with (x, y) coordinates into three distinct clusters. The given points are A1(2,10), A2(2,5), A3(8,4), B1(5,8), B2(7,5), B3(6,4), C1(1,2), and C2(4,9). Using the Euclidean distance function, demonstrate the iterative process of clustering by initially selecting A1(2,10), B1(5,8), and C1(1,2) as the initial centroids. Show the step-by-step reassignment of points to clusters, centroid updates, and the final cluster formation once convergence is reached.
- b) What is reinforcement learning? Explain how nondeterministic rewards influence an agent's learning process.

- a) Explain the importance of model evaluation in machine learning. Discuss key classification metrics (Confusion Matrix, Accuracy, Precision, Recall, F1 Score, AUC-ROC Curve) and regression metrics (MSE, RMSE, MAE, RMSLE), highlighting their significance, differences, and appropriate use cases.
- b) Apply the agglomerative hierarchical clustering algorithm to the given one-dimensional data points: 18, 22, 25, 42, 27, and 43. Using the minimum distance (single linkage) method, construct the hierarchical clustering dendrogram. At each iteration, merge the closest clusters, update the proximity matrix, and clearly present each step of the clustering process until a single cluster remains.

8 L2 5 2

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BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
